From Threshold Belief to Epistemic Phase Geometry: Mechanized Dynamics in a Four-Valued Probabilistic Epistemic Logic

Julian L. Voigt (cr4bbz)

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Abstract

I investigate how four-valued evidence changes under probability-only conditionalization and what is lost when quantitative support is reduced to categorical belief. In the 4-PEL semantics, belief thresholds positive and negative support separately. A primitive, qualitative operator called evidence-stable knowledge returns that belief value on homogeneous accessible profiles and strict falsity otherwise. The term "knowledge" denotes this stipulated operator, not an independently established theory of epistemic justification.

A six-world witness family realizes all sixteen source–target values of $\mathsf{K}(\mathsf{B}p)$. For a chosen rational affine interpolation of support endpoints, threshold-bit changes determine unique wall-hit parameters; their order determines adjacent intermediate states. These parameters describe the interpolation, not an inferred history of the discrete update. Separately, normalized nonnegative world weights yield certified finite measures and convex model-valued paths whose fixed-event masses are affine. Formula-level identification for probability-sensitive events and generation of intermediate models by conditionalization remain open.

A further six-cell refinement records reliability tags while preserving four-cell support and convex mixing. Fine modal stability is equivalent to coarse stability plus local fibre rigidity; explicit S5-frame counterexamples show that coarse stability alone loses reliability information. This is a semantic projection result, not an axiomatization of a six-valued logic.

The Lean 4.31 artifact distinguishes the lightweight model interface from probability-integrity-certified models and records declaration-level axiom dependencies, rejecting native evaluation in the selected manuscript proof chains. The contribution is a mechanized case study with explicit preservation and information-loss boundaries; no completeness, general continuity, or priority claim is made.

1 Introduction

Probabilistic epistemic models are usually asked two different questions. One concerns degrees: how much support does an agent assign to a proposition? The other concerns categories: does the agent believe, know, reject, or remain undecided? Threshold approaches connect the two by turning sufficiently high probability into all-or-nothing belief. Four-valued semantics adds a second axis. Positive and negative support can coexist or both fail, so the categorical result is no longer binary but belongs to

$$\{\mathbf{T}, \mathbf{F}, \mathbf{B}, \mathbf{N}\} = \{(1,0), (0,1), (1,1), (0,0)\},$$

following the Belnap–Dunn/FDE tradition [1, 5].

The 4-PEL project combines these ideas in a finite-accessibility evidence semantics. Positive and negative extensions are measured separately by the same local set function. A Lockean threshold is applied to each mass, producing a four-valued belief state. A separate evidence-stable knowledge operator then checks whether the complete FDE profile is invariant across the accessible range. This yields an exact factorization:

$$\mathsf{K}_i\varphi = \begin{cases} \mathsf{B}_i\varphi, & \mathsf{Stable}_i(\varphi), \\ \mathbf{F}, & \text{otherwise.} \end{cases}$$

The knowledge operator is therefore not a second probability threshold. It is a qualitative stability filter on top of threshold belief.

The lightweight interface and its stronger probability-certified subclass are distinguished in Section 2. The paper studies one-step conditionalization and, separately, a chosen affine interpolation of its support endpoints:

Research Question 1.1 (Dynamic phase geometry)**.** Which four-valued belief and knowledge endpoints can admissible conditionalization connect, and what additional phase structure follows if their signed supports are interpolated affinely?

The machine-checked answer develops in several layers. First, admissible conditionalization can both destroy and restore the stability gate. More strongly, a fixed six-world construction scheme realizes every ordered pair of four-valued values at the level of $\mathsf{K}(\mathsf{B}p)$. Thus the categorical reachability graph is complete. That global permissiveness is paired with a sharp local invariant: a belief value changes exactly when at least one of its two threshold decisions changes.

The four categorical states can consequently be treated as the vertices of a Boolean square. The number of changed threshold coordinates defines a wall count

$$d_{\text{thr}} \in \{0, 1, 2\}.$$

Zero walls means the same FDE state, one wall means an edge move, and two walls mean a diagonal move. The two diagonals are $\mathbf{T} \leftrightarrow \mathbf{F}$ and $\mathbf{N} \leftrightarrow \mathbf{B}$.

The geometric interpretation can then be reconnected to the underlying rational support masses. A changed threshold bit is exactly endpoint straddling of the Lockean boundary. For the explicit affine interpolation

$$\gamma(t) = x + t(y - x), \qquad t \in \mathbb{Q},$$

straddling produces a rational crossing parameter $t \in [0, 1]$ at which $\gamma(t) = c$. A two-wall transition therefore contains two threshold events, one for positive support and one for negative support. Each affine crossing time is unique. Their order is positive-first, simultaneous, or negative-first, and the simultaneous case is exactly passage through the point (c, c) in the support plane.

If the two crossings are not simultaneous, the rational midpoint between them occupies a categorical state adjacent to both diagonal endpoints. The order of the wall events determines the intermediate state. For example,

$$\mathbf{N} \to \mathbf{B}$$

becomes either

$$\mathbf{N} \to \mathbf{T} \to \mathbf{B}$$

or

$$\mathbf{N} \to \mathbf{F} \to \mathbf{B}.$$

The same classification holds for the other diagonal orientations. Here "time" means the interpolation parameter, not elapsed time or an inferred intermediate stage of a one-step update. Wall count counts differing endpoint bits, not crossings along an arbitrary path.

A separate verified layer constructs complete probability-integrity-certified models. Nonnegative normalized rational point weights generate local measures satisfying an explicit finite-probability integrity contract. Convex interpolation preserves those distributions and makes the mass of every fixed event affine in the interpolation parameter. The local paths can then be packaged into complete strong 4-PEL models while worlds, accessibility, atomic valuation, and threshold remain fixed.

For fixed events this yields a compact structural picture; applying it to the support of a varying modal formula requires an additional bridge:

$$\boxed{\begin{array}{c}\text{probability simplex} \longrightarrow \text{strong model path}\\ \longrightarrow \text{affine fixed-event masses} \longrightarrow \text{threshold sides}\\ \longrightarrow \text{wall crossings} \longrightarrow \text{crossing order}\\ \longrightarrow \text{FDE phase sequence.}\end{array}}$$

Reliability refinement. Section 13 studies a second information-loss map: six reliability-tagged evidence cells project to four coarse cells. Projection preserves support and mixing, but not the recoverability of reliability. Fine modal stability is equivalent to coarse stability plus local fibre rigidity. These elementary semantic results are not a six-valued logical calculus; they extend the paper's investigation of which structure a coarse observation forgets.

Methodological discipline. The paper distinguishes four levels throughout. Machine-checked theorems are reported as such. Finite witness constructions are not silently generalized beyond their quantified statement. Geometric and philosophical readings are identified as interpretations. Literature and novelty claims remain deliberately modest until a systematic comparison has been completed.

Current boundary. Model-valued convex paths are now verified for weight-generated strong endpoint models on a common semantic skeleton. This does not yet imply that every legacy conditionalization witness has such a strong representation, that every intermediate point is itself generated by conditionalization of the source model, or that support events defined by arbitrary probability-sensitive modal formulas remain fixed along the path. The rational affine crossing theorem also remains distinct from an abstract intermediate-value theorem over arbitrary continuous paths. These formula-level, update-level, and topological questions are the next mathematical boundaries.

2 Four-Valued Probabilistic Epistemic Semantics

The formal development starts with a lightweight, finite-accessibility interface

$$M = (W, R, \mu, v, c),$$

where W is the world type, $R_i(w)$ is a finite accessibility list, $\mu_{i,w}$ is a rational-valued function on event lists, v assigns FDE values to atoms, and c_i satisfies

$$\frac{1}{2} < c_i \leq 1.$$

The interface requires only $\mu_{i,w}(R_i(w)) = 1$ and $\mu_{i,w}(\varnothing) = 0$, besides the threshold bounds. It does *not* require nonnegativity, additivity, extensionality, or duplicate-free lists. Consequently a general `Model` is not yet a certified probability model; "support mass" below means its rational set-function value unless integrity is explicitly assumed. The stronger `StrongProbabilityMode l` contract in Section 9 supplies ordinary finite-measure laws on accessible, duplicate-free events. Both signed masses then lie in $[0, 1]$.

The stored `worlds` list need not exhaust the type W, and the interface does not enforce accessibility closure into that list. The displayed finite witnesses do use exhaustive finite types and closed ranges. Generic theorems quantify over the literal Lean interface, not over an unstated finite-universe axiom. No reflexivity, transitivity, or Euclidean condition is generally assumed. Normalization does exclude an empty accessible range: otherwise its mass would be both zero and one (`PaperReview.accessibility_nonempty`).

2.1 Bilateral four-valued information

A semantic value is a pair of Boolean coordinates

$$x = (x^+, x^-),$$

with the four canonical values

$$\mathbf{T} = (1,0), \qquad \mathbf{F} = (0,1), \qquad \mathbf{B} = (1,1), \qquad \mathbf{N} = (0,0).$$

The positive coordinate records support for the formula and the negative coordinate records support against it. Thus $\mathbf{B}$ is a glut and $\mathbf{N}$ is a gap. Negation exchanges the two coordinates, while conjunction takes conjunction on positive support and disjunction on negative support. Independence here means that the coordinates are not logical complements; it does not assert stochastic independence of their support events. In an integrity-certified model the two events share one measure and may overlap.

This bilateral representation is crucial for the dynamic analysis. The two coordinates can cross a probabilistic threshold at different times, producing a temporal structure that would be invisible in a single Boolean belief status.

2.2 Threshold belief

For a modal formula φ, define its positive and negative accessible extensions at (i, w) by

$$E^+_{i,w}(\varphi) = \{u \in R_i(w) : [\![\varphi]\!]^+_u = 1\},$$

$$E^-_{i,w}(\varphi) = \{u \in R_i(w) : [\![\varphi]\!]^-_u = 1\}.$$

The corresponding support masses are

$$P^+_{i,w}(\varphi) = \mu_{i,w}(E^+_{i,w}(\varphi)), \qquad P^-_{i,w}(\varphi) = \mu_{i,w}(E^-_{i,w}(\varphi)).$$

Probabilistic belief thresholds the two masses independently:

$$\mathsf{B}_i\varphi = \Big([P^+_{i,w}(\varphi) \geq c_i], [P^-_{i,w}(\varphi) \geq c_i]\Big).$$

Accordingly, $\mathsf{B}_i\varphi = \mathbf{B}$ is possible when both signed support masses cross the threshold, while $\mathsf{B}_i\varphi = \mathbf{N}$ occurs when neither does.

2.3 Evidence-stable knowledge

The modal object language contains primitive probabilistic belief B, primitive evidence-stable knowledge K, and primitive raw accessibility possibility. The latter is kept distinct from the internal expression $\neg\mathsf{K}\neg\varphi$, because the two need not agree in a four-valued setting.

For the present paper, the central modal predicate is full-value stability. Let

$$\mathsf{Stable}^M_i(w, \varphi)$$

mean pairwise homogeneity on the accessible range:

$$H_i(w, \varphi) \iff \forall u, z \in R_i(w), \quad [\![\varphi]\!]_u = [\![\varphi]\!]_z.$$

Here $H_i = \mathsf{Stable}_i^M$. This does not compare the range with the current world unless that world is accessible. The primitive definition, writing $x_u = [\![\varphi]\!]_u$, is

$$(\mathsf{K}_i\varphi)^+ = H_i \wedge \forall u \in R_i(w)\; x_u^+, \qquad (\mathsf{K}_i\varphi)^- = \neg H_i \vee \exists u \in R_i(w)\; x_u^-.$$

These are classical meta-level Boolean operations on evidence coordinates. In particular, primitive K does not directly inspect μ or c. Heterogeneity is stipulated to count as negative evidence for this operator. Without frame assumptions no factivity principle at the current world is inferred; even on reflexive frames, a glut is not classical truth only.

The resulting relationship between knowledge and belief is exact.

Theorem 2.1 (Knowledge as stability-filtered belief)**.** *For every finite 4-PEL model, agent, world, and modal formula,*

$$\mathsf{K}_i\varphi = \begin{cases} \mathsf{B}_i\varphi, & \mathsf{Stable}_i(\varphi), \\ \mathbf{F}, & \neg\mathsf{Stable}_i(\varphi). \end{cases}$$

STATUS: PROVED; TRUST INVENTORY IN APPENDIX A.

Proof sketch. The accessible range is nonempty. If homogeneous, each signed extension is either the entire range or empty. Normalization and $0 < c_i \leq 1$ make its threshold bit equal the corresponding common evidence bit. The definition of K returns those same bits. If heterogeneous, its positive bit is false and its negative bit true. No additivity assumption is needed. $\square$

Hence an unstable accessible profile is not merely low-confidence evidence. It is a structural failure mode that absorbs every probabilistic belief value into strict $\mathbf{F}$ at the knowledge layer. Conversely, on a homogeneous profile the stability filter is transparent and $\mathsf{K}_i\varphi$ preserves the complete four-valued belief state.

For a fixed interpretation $u \mapsto x_u$ and fixed accessible range, changing only probability cannot change K. This is verified separately by `PaperReview.knowledge_ignores_probability_for_fixed_values`. Probability affects $\mathsf{K}\varphi$ only through probability-sensitive subformulas of φ; the two-case factorization is not a claim of two independent numerical inputs to primitive knowledge.

3 Admissible Conditionalization and Dynamic Knowledge Phases

The update operator conditions each local set function on the prior positive extension of an evidence formula E in the legacy language (atoms, negation, conjunction, and belief). It is not currently defined for arbitrary formulas with primitive K or raw possibility. Let

$$E_{i,w}^+ = \{u \in R_i(w) : [\![E]\!]_u^+ = 1\}.$$

For a finite set $S \subseteq W$, the raw updated local probability is

$$\mu_{i,w}^E(S) = \frac{\mu_{i,w}(S \cap E_{i,w}^+)}{\mu_{i,w}(E_{i,w}^+)}$$

whenever the denominator is nonzero. At the model level, the Lean constructor requires an explicit *ConditionalizationAdmissible* proof. Its field named `positive_mass` literally asserts *nonzero* mass at every agent/world pair, not strict positivity. Nonzero becomes positive when nonnegativity is separately available. Two further proof fields require the updated total mass to be one and its empty-event mass zero. They do not certify all strong probability laws. The event is evaluated in the prior model and held fixed for this update; this is not iterative reevaluation of its formula.

The update is intentionally conservative:

$$W^E = W, \qquad R^E = R, \qquad v^E = v, \qquad c^E = c,$$

and only μ changes. Thus any dynamic change in a formula must ultimately be mediated by a semantic dependency on probability.

3.1 A syntactic invariant sector

The modal language has a probability-free fragment containing atoms, negation, conjunction, primitive knowledge, and raw possibility, but no occurrence of the probabilistic belief constructor. Lean proves that every such formula preserves its complete FDE value under admissible conditionalization. Its accessible stability profile, outer knowledge, and raw possibility are therefore preserved as well.

This result isolates a first dynamic boundary:

$$\text{no probabilistic belief occurrence} \quad \Longrightarrow \quad \text{complete value invariant under conditionalization.}$$

The converse is false. Later sections exhibit formulas containing B whose probabilities change while their threshold status remains invariant.

3.2 Two cases in the knowledge factorization

The factorization theorem from Section 2 yields four combinations of the before/after stability bits for $\mathsf{K}_i\varphi$. Let M_0 and M_1 be two models on the same modal language. The stability states of φ determine how the belief channel is exposed at the knowledge layer:

Stable_{M_0}	Stable_{M_1}	knowledge phase
true	true	$\mathsf{K} = \mathsf{B}$ on both sides
true	false	posterior $\mathsf{K} = \mathbf{F}$
false	true	prior $\mathsf{K} = \mathbf{F}$, posterior $\mathsf{K} = \mathsf{B}$
false	false	$\mathsf{K} = \mathbf{F}$ on both sides

The classification is update-independent: it follows directly from the pointwise factorization. The conditionalization development additionally contains finite witnesses for both off-diagonal phases. One update fractures a homogeneous $\mathsf{B}p$ profile into a heterogeneous profile, giving

$$\mathsf{Stable}(\mathsf{B}p): \text{ true} \to \text{false}, \qquad \mathsf{K}(\mathsf{B}p): \ \mathbf{T} \to \mathbf{F}.$$

A second witness restores stability in the opposite direction:

$$\mathsf{Stable}(\mathsf{B}p): \text{ false} \to \text{true}, \qquad \mathsf{K}(\mathsf{B}p): \ \mathbf{F} \to \mathbf{T}.$$

Corollary 3.1 (Truth-only status can be lost and regained)**.** *There are admissible conditionalizations taking* $\mathsf{K}(\mathsf{B}p)$ *from* $\mathbf{T}$ *to* $\mathbf{F}$*, and others from* $\mathbf{F}$ *to* $\mathbf{T}$*. Thus preservation of the truth-only status is not an unconditional update law. This is not a monotonicity theorem about unspecified truth or information orders.* STATUS: FINITE-WITNESS; TRUST INVENTORY IN APPENDIX A.

This phase table motivates the stronger question addressed next. Instead of asking only whether stability can fracture or recover, one can ask which complete four-valued endpoints are reachable when the belief profile is kept homogeneous so that the outer knowledge filter remains transparent.

4 Complete Dynamic Epistemic Reachability

The strongest endpoint theorem uses a uniform six-world construction scheme. For arbitrary source and target values

$$s, t \in \{\mathbf{T}, \mathbf{F}, \mathbf{B}, \mathbf{N}\},$$

consider worlds

$$\{f, a_1, a_2, a_3, a_4, r\}.$$

The distinguished focus world f carries the target value t for atom p, each of the four anchor worlds a_k carries the source value s, and the spare world r carries $\mathbf{N}$. A second atom e is $\mathbf{T}$ only at f and $\mathbf{F}$ elsewhere. Every world accesses all six worlds. The prior measure is uniform,

$$\mu(\{w\}) = \frac{1}{6},$$

and the Lockean threshold is

$$c = \frac{2}{3}.$$

Before update, the four anchors contribute total mass $4/6 = 2/3$. Coordinate by coordinate, they therefore force the threshold belief $\mathsf{B}p$ to reproduce the source value s, regardless of the single target-bearing focus world. After conditioning on e, all posterior mass is concentrated at f, so $\mathsf{B}p$ reproduces the target value t.

The local probability function is independent of the evaluation world. Consequently, the complete $\mathsf{B}p$ profile is homogeneous across the common accessible range before and after update. The outer knowledge operator therefore passes both values through unchanged.

Theorem 4.1 (Complete dynamic epistemic reachability)**.** *For every ordered pair (s, t) of complete FDE values, there is a member of the six-world family and an admissible conditionalization such that*

$$\mathsf{K}(\mathsf{B}p) : s \longrightarrow t.$$

Across that update the accessibility relation, atomic valuation, and threshold are fixed; only the local probability field changes. STATUS: FINITE-WITNESS; TRUST INVENTORY IN APPENDIX A.

Proof sketch. For either sign, a true source bit contributes at least $4/6 = c$, while a false source bit receives at most $1/6 < c$ from the focus. The evidence has mass $1/6$; conditioning gives focus mass one. Homogeneity of the inner belief value makes each of its outer signed support events either the whole range or empty, so $\mathsf{K}(\mathsf{B}p)$ has the asserted endpoints. The source bits equal to the threshold use the inclusive $\geq$ convention. $\square$

Equivalently, the categorical reachability graph is complete:

source \ target	**T**	**F**	**B**	**N**
T	✓	✓	✓	✓
F	✓	✓	✓	✓
B	✓	✓	✓	✓
N	✓	✓	✓	✓

The theorem immediately gives creation and removal of both nonclassical statuses. In particular, the development contains verified instances

$$\mathbf{T} \to \mathbf{B}, \qquad \mathbf{B} \to \mathbf{N}, \qquad \mathbf{N} \to \mathbf{T}.$$

Thus neither glutty nor gappy knowledge is dynamically terminal under the present update semantics.

Quantifier discipline. The theorem does not say that one fixed concrete model realizes all sixteen arrows. The family is parameterized by the requested source and target values. For each pair, however, the pre-update and post-update models share the same R, v, and c, and the transition is generated by admissible probability-only conditionalization. The result is therefore a semantic non-prohibition theorem: no ordered pair of complete four-valued knowledge statuses is ruled out merely by the conditionalization mechanism.

The uniform and point-mass measures have the intended finite-probability reading, but the current Lean reachability statement returns the lightweight `Model`, not a `StrongProbabilityModel` certificate. Its separate strong-promotion obligation is not discharged by the generic convex-path constructor.

The reachability theorem establishes permissiveness across the witness family. The next result identifies the exact local condition under which no categorical change occurs.

5 Threshold-Side Robustness and the Boolean Square

Complete reachability shows how permissive the dynamics can be across different models in the witness family. The complementary question is local: what must change in order for one probabilistic belief value to change at all?

Let

$$p^+ = P^+_{i,w}(\varphi), \qquad p^- = P^-_{i,w}(\varphi).$$

Then the complete belief value is exactly the pair of threshold decisions

$$\mathsf{B}_i\varphi = ([p^+ \geq c_i], [p^- \geq c_i]).$$

This gives an immediate semantic boundary.

Theorem 5.1 (Threshold-side robustness)**.** *For any two models on the same language, at a fixed agent, world, and formula,*

$$\mathsf{B}_i^{\text{after}}\varphi = \mathsf{B}_i^{\text{before}}\varphi$$

if and only if the positive threshold decision is unchanged and the negative threshold decision is unchanged. Status: Proved; trust inventory in Appendix A.

This equivalence is immediate from equality of Boolean pairs; it is a useful interface lemma, not a new probability law. In the general two-model theorem, each model uses its own threshold. Under conditionalization the threshold is fixed. The raw masses may therefore move substantially while the categorical result remains fixed. The invariant is not the precise support mass but the pair of threshold sides occupied by the two masses.

The Lean development gives a sufficient compositional certificate (not a complete characterization of invariance for all formulas), `ModalConditionalizationRobust`. At a belief node, robustness requires only that the two threshold decisions be preserved at every world. The embedded formula itself need not be dynamically invariant. Negation, conjunction, evidence-stable knowledge, and raw possibility propagate robustness compositionally. The previously verified probability-free fragment is contained in this semantic class, but the inclusion is strict: a diagonal member of the reachability family gives a formula containing B whose probability distribution changes nontrivially while the belief value stays fixed.

5.1 The threshold square

Since every FDE value is already a pair of threshold bits, the categorical state space is the Boolean square

$$\mathbf{N} = (0,0), \qquad \mathbf{T} = (1,0), \qquad \mathbf{F} = (0,1), \qquad \mathbf{B} = (1,1).$$

$$\begin{tikzcd}
(0,1) & (1,1) \\
\mathbf{F} \arrow[r, dash] \arrow[d, dash] \arrow[dr, dash, dashed] & \mathbf{B} \arrow[d, dash] \arrow[dl, dash, dashed] \\
\mathbf{N} \arrow[r, dash] & \mathbf{T} \\
(0,0) & (1,0)
\end{tikzcd}$$

The solid edges indicate undirected adjacency, not a preferred update direction or a factivity ordering. Define the endpoint threshold-wall count

$$d_{\text{thr}}(a,b) = \mathbf{1}[a^+ \neq b^+] + \mathbf{1}[a^- \neq b^-].$$

Lean verifies the basic geometry:

$$0 \leq d_{\text{thr}}(a,b) \leq 2,$$

$$d_{\text{thr}}(a,b) = 0 \iff a = b,$$

and

$$d_{\text{thr}}(a,b) = 2 \iff a^+ \neq b^+ \;\wedge\; a^- \neq b^-.$$

Thus one-wall changes are exactly edge moves and two-wall changes are exactly diagonal moves. The two diagonal pairs are

$$\mathbf{N} \leftrightarrow \mathbf{B}, \qquad \mathbf{T} \leftrightarrow \mathbf{F}.$$

Proposition 5.2 (Robustness is zero displacement)**.** *Every compositionally robust modal formula has threshold-wall count zero under its designated admissible conditionalization.* Status: Proved; trust inventory in Appendix A.

The reachability construction realizes every displacement class. Hence the dynamic picture combines two properties that might initially look opposed:

global categorical reachability is complete,

while

local categorical change is exactly coordinatewise threshold change.

This tension is the starting point for the geometric refinement in the next sections.

6 From Threshold Straddling to Literal Affine Crossings

The Boolean square records which threshold decisions differ, but it does not yet say what numerical event lies behind a changed bit. The next formal layer reconnects categorical displacement to rational support masses.

6.1 Endpoint straddling

For a rational threshold c, define

$$\mathrm{Straddle}(c; x, y)$$

to mean that exactly one endpoint lies on or above the threshold:

$$(c \leq x \wedge \neg(c \leq y)) \;\vee\; (c \leq y \wedge \neg(c \leq x)).$$

For rational numbers this is the expected endpoint pattern

$$x < c \leq y \quad \text{or} \quad y < c \leq x.$$

Theorem 6.1 (Threshold decision versus endpoint straddling)**.** *For rational c, x, y,*

$$[c \le x] \ne [c \le y] \iff \mathrm{Straddle}(c; x, y).$$

STATUS: PROVED; TRUST INVENTORY IN APPENDIX A.

At belief level, the total wall count decomposes into the one-dimensional wall counts of the positive and negative support coordinates. Lean therefore verifies the exact three-way correspondence:

$$\begin{aligned} d_{\mathrm{thr}} = 0 &\iff \text{neither support coordinate straddles,} \\ d_{\mathrm{thr}} = 1 &\iff \text{exactly one support coordinate straddles,} \\ d_{\mathrm{thr}} = 2 &\iff \text{both support coordinates straddle.} \end{aligned}$$

In particular, every genuine categorical belief change has a numerical endpoint witness on at least one signed support axis.

6.2 Affine interpolation

Endpoint straddling still describes only two snapshots. To expose an actual crossing event, the formalization introduces the directed rational affine path

$$\gamma_{x,y}(t) = x + t(y - x), \qquad t \in \mathbb{Q}.$$

It satisfies

$$\gamma_{x,y}(0) = x, \qquad \gamma_{x,y}(1) = y.$$

For the increasing orientation $x < c \le y$, the explicit threshold parameter is

$$t_+ = \frac{c - x}{y - x}.$$

For the decreasing orientation $y < c \le x$, the development uses the equivalent directed parameter

$$t_- = \frac{x - c}{x - y}.$$

Lean Core rational arithmetic is sufficient to prove that the relevant parameter lies in the unit interval and hits the threshold exactly.

Theorem 6.2 (Affine unit-interval threshold crossing)**.** *If rational endpoints x, y straddle rational threshold c, then*

$$\exists t \in \mathbb{Q} : 0 \le t \le 1 \quad \wedge \quad \gamma_{x,y}(t) = c.$$

STATUS: PROVED; TRUST INVENTORY IN APPENDIX A.

Proof sketch. Straddling implies distinct endpoints. Solve $x + t(y - x) = c$ for $t = (c - x)/(y - x)$; the straddling inequalities give $0 \le t \le 1$, accounting for the sign of the denominator. Substitution verifies the hit. A hit can be at an endpoint; the theorem does not require $0 < t < 1$. □

No abstract topology is used, and there is no Mathlib dependency.

Corollary 6.3 (Belief change has a literal affine wall hit)**.** *If admissible conditionalization changes the complete FDE value of $\mathsf{B}_i\varphi$, then at least one of the two chosen interpolations of its endpoint support masses*

$$t \mapsto P_t^+(\varphi), \qquad t \mapsto P_t^-(\varphi)$$

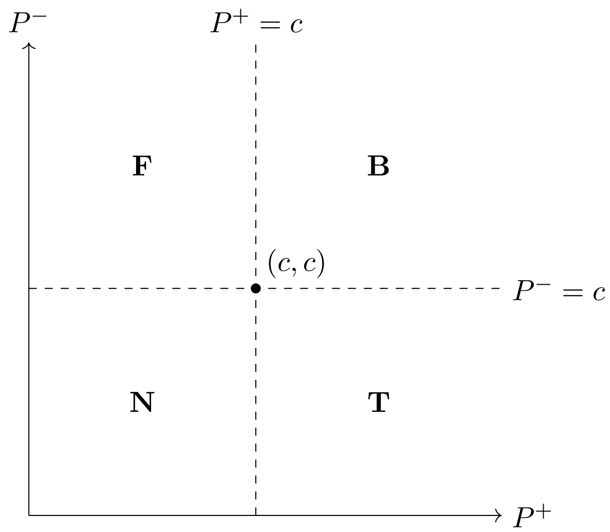
hits c_i at some rational $t \in [0, 1]$. Here $P_t^+ = (1 - t)P_0^+ + tP_1^+$, and likewise for negative support; this notation does not yet denote evaluation in an intermediate model. STATUS: PROVED; TRUST INVENTORY IN APPENDIX A.

For a two-wall transition, both signed support coordinates have crossing witnesses, though they need not cross at the same parameter. This asks an order question about the chosen interpolation: which wall is hit first?

Support-plane picture. The two masses define a point

$$(P^+, P^-) \in \mathbb{Q}^2.$$

The threshold lines $P^+ = c$ and $P^- = c$ partition the plane into the four FDE regions.



The threshold square is the four-region categorical observation of the signed support plane. These four regions must not be confused with the four evidence masses (t, b, n, f) in a probability simplex. For integrity-certified models the relevant support points lie in the unit square; the algebraic lemmas themselves permit arbitrary rational coordinates.

7 Unique Crossing Times and Temporal Order

A two-wall transition gives one affine threshold hit on the positive support coordinate and one on the negative coordinate. To turn these hits into a temporal classification, the crossing parameter must be intrinsic rather than an arbitrary existential witness.

7.1 Uniqueness

Suppose $x \neq y$. The affine map

$$\gamma_{x,y}(t) = x + t(y - x)$$

is injective in its parameter. Lean proves this directly over the rationals by cancelling the nonzero factor $y - x$.

Theorem 7.1 (Unique affine threshold time)**.** *Let $x \neq y$. If*

$$\gamma_{x,y}(t) = c \qquad \textit{and} \qquad \gamma_{x,y}(s) = c,$$

then

$$t = s.$$

Consequently every straddling affine segment has a unique threshold crossing parameter in $[0, 1] \cap \mathbb{Q}$.
Status: Proved; trust inventory in Appendix A.

For a two-wall belief transition write

$$t_+ = \text{the unique positive-support crossing time},$$

$$t_- = \text{the unique negative-support crossing time}.$$

Both lie in the rational unit interval.

7.2 Crossing-order trichotomy

The linear order on $\mathbb{Q}$ now yields an exhaustive classification:

$$t_+ < t_-, \qquad t_+ = t_-, \qquad t_- < t_+.$$

Theorem 7.2 (Affine crossing-order trichotomy)**.** *Every two-wall conditionalized belief transition admits intrinsic positive and negative crossing times satisfying exactly one of*

$$t_+ < t_-, \qquad t_+ = t_-, \qquad t_- < t_+.$$

STATUS: PROVED; TRUST INVENTORY IN APPENDIX A.

The three cases have immediate event readings:

$$\begin{aligned} t_+ < t_- &\ :\ \text{positive threshold wall first}, \\ t_+ = t_- &\ :\ \text{simultaneous crossing}, \\ t_- < t_+ &\ :\ \text{negative threshold wall first}. \end{aligned}$$

7.3 The simultaneous case

The equality case has a geometric characterization stronger than mere coincidence of two stored parameters.

Theorem 7.3 (Simultaneity as the wall intersection)**.** *For a two-coordinate affine crossing pair,*

$$t_+ = t_-$$

if and only if there exists one rational parameter t such that

$$P_t^+ = c \qquad \textit{and} \qquad P_t^- = c.$$

Equivalently, the support path passes through the wall-intersection point

$$(c, c).$$

STATUS: PROVED; TRUST INVENTORY IN APPENDIX A.

Thus simultaneity requires the explicit equality of the two crossing parameters. No probability-zero or genericity theorem about this condition is claimed. If $t_+ \neq t_-$, there is a nonempty rational interval between the two wall events. The next section proves that this interval carries a forced categorical phase.

Why order matters. Categorical endpoint values alone do not specify which threshold coordinate changes first along a chosen affine segment. Yet for a diagonal transition both bits must change. The temporal order of those changes therefore contains information absent from the pair $(\text{source}, \text{target})$. Crossing-order geometry recovers exactly that lost information.

8 Intermediate FDE Phases Between Sequential Crossings

Suppose a two-wall transition is not simultaneous. The two unique crossing times then determine an open interval in which exactly one threshold coordinate has already switched to its target side.

The formal development chooses the rational midpoint

$$t_m = \frac{t_+ + t_-}{2}.$$

If $t_+ < t_-$, Lean proves

$$t_+ < t_m < t_-.$$

If $t_- < t_+$, the symmetric inequalities hold. These midpoint facts require only rational arithmetic.

8.1 Threshold side before and after a crossing

The affine support coordinate is strictly monotone whenever its endpoints are distinct. For an increasing straddling path, parameters before the unique crossing remain below threshold and parameters after it lie above threshold. For a decreasing path, the threshold sides reverse. The formal result can be stated without choosing an orientation:

Proposition 8.1 (Source side before, target side after)**.** *Let a rational affine support coordinate straddle c and hit c uniquely at time t_c. Then*

$$t < t_c \quad \Longrightarrow \quad \mathrm{side}_c(\gamma(t)) = \mathrm{side}_c(x),$$

and

$$t_c < t \quad \Longrightarrow \quad \mathrm{side}_c(\gamma(t)) = \mathrm{side}_c(y).$$

Status: Proved; trust inventory in Appendix A.

This is the key bridge from event order to a categorical intermediate state.

8.2 Generic intermediate vertices

Let

$$s = (s^+, s^-), \qquad t = (t^+, t^-)$$

be opposite vertices of the threshold square. Define

$$I_+(s,t) = (t^+, s^-)$$

for positive-first motion and

$$I_-(s,t) = (s^+, t^-)$$

for negative-first motion.

Theorem 8.2 (Positive-first midpoint phase)**.** *If both support coordinates straddle and $t_+ < t_-$, then at the rational midpoint*

$$\mathit{state}(t_m) = I_+(s,t).$$

That is, the positive coordinate has already reached the target threshold side while the negative coordinate is still on the source threshold side. Status: Proved; trust inventory in Appendix A.

Theorem 8.3 (Negative-first midpoint phase)**.** *If both support coordinates straddle and $t_- < t_+$, then*

$$\mathit{state}(t_m) = I_-(s,t).$$

Status: Proved; trust inventory in Appendix A.

For opposite endpoints, both intermediate states are adjacent to source and target:

$$d_{\mathrm{thr}}(s, I_\pm(s,t)) = 1, \qquad d_{\mathrm{thr}}(I_\pm(s,t), t) = 1.$$

Hence every sequential diagonal displacement decomposes into two one-wall phases.

Boundary values at a simultaneous hit. The threshold convention is inclusive. At the common hit (c,c) the value is always **B**, verified by `PaperReview.threshold_intersection_is_B`. Consequently "no intermediate open phase" does not mean "no intermediate value at any parameter." For example, at $c = 3/5$, the segment from $(4/5, 2/5)$ to $(2/5, 4/5)$ has **T** at zero, **B** at $1/2$, and **F** at one. The common hit is an isolated **B** between the two open phases; `PaperReview.simultaneous_T_F_has_B_hit` checks the three values. The theorem about a sequential midpoint does not include the crossing instants.

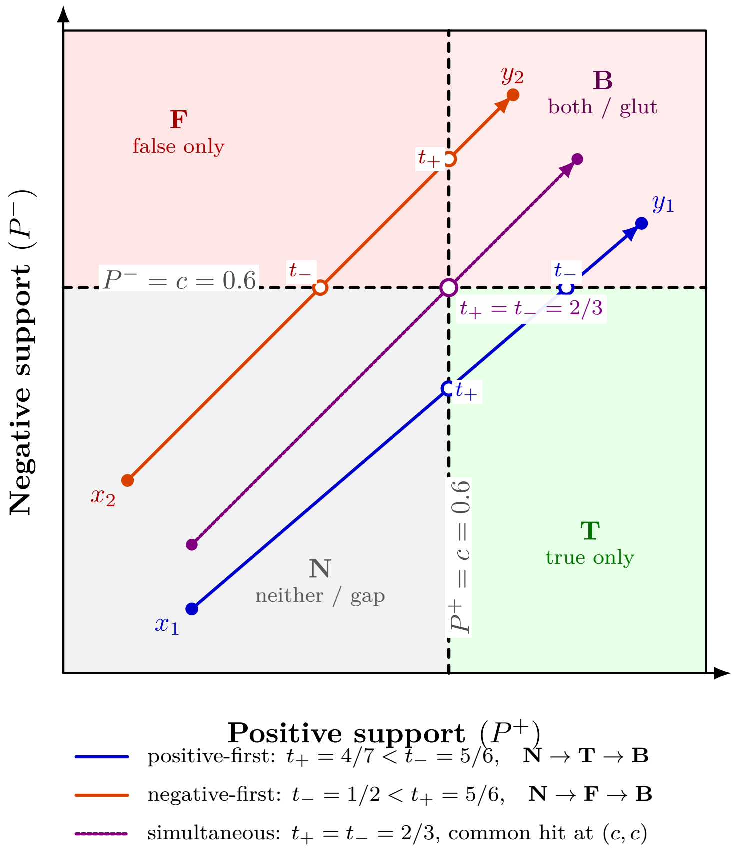


Figure 1: Affine phase geometry for three $\mathbf{N} \to \mathbf{B}$ support trajectories at threshold $c = 0.6$. Positive-first and negative-first paths necessarily occupy $\mathbf{T}$ and $\mathbf{F}$, respectively, between their unique wall-crossing times. The simultaneous path is the exceptional case $t_+ = t_-$ and passes through the wall intersection (c, c). The figure visualizes the Lean-verified crossing-order and intermediate-phase theorems; it does not by itself assert that every interpolated support point is already realized by a full admissible model.

8.3 Concrete diagonal table

The complete verified table is:

Diagonal transition	positive wall first	negative wall first
$\mathbf{N} \to \mathbf{B}$	$\mathbf{T}$	$\mathbf{F}$
$\mathbf{B} \to \mathbf{N}$	$\mathbf{F}$	$\mathbf{T}$
$\mathbf{T} \to \mathbf{F}$	$\mathbf{N}$	$\mathbf{B}$
$\mathbf{F} \to \mathbf{T}$	$\mathbf{B}$	$\mathbf{N}$

Thus, for example,

$$\mathbf{N} \to \mathbf{B}$$

can have either phase sequence

$$\mathbf{N} \to \mathbf{T} \to \mathbf{B}$$

or

$$\mathbf{N} \to \mathbf{F} \to \mathbf{B},$$

and the choice is determined exactly by whether positive or negative support crosses the Lockean wall first.

Figure 1 makes the trichotomy explicit for three rational affine trajectories with the same threshold. The two sequential cases pass through different adjacent FDE cells, whereas the simultaneous path hits the unique wall intersection and therefore has no open interval carrying an intermediate adjacent phase.

Theorem 8.4 (Two-wall intermediate-phase classification)**.** *Every two-wall conditionalized belief transition admits an affine crossing pair such that either*

(i) the two crossing times are equal and the support path hits (c, c), or

(ii) the positive crossing is earlier and the midpoint has the positive-first intermediate FDE state, or

(iii) the negative crossing is earlier and the midpoint has the negative-first intermediate FDE state.

Status: Proved; trust inventory in Appendix A.

Scope of this theorem. The theorem in this section classifies the explicit affine interpolation of the two support masses. Section 9 subsequently verifies a genuine path of complete strong 4-PEL models for weight-generated endpoints and proves that every fixed event mass follows exactly such an affine trajectory. This does not realize arbitrary legacy support endpoints as a strong model path. For weight-generated endpoints the further gap is the formula-level identification of the positive and negative support events with fixed events along the model path. For arbitrary probability-sensitive modal formulas those event memberships may themselves change with the path parameter.

9 From Probability Simplexes to Strong Model Paths

The affine crossing results of the previous sections are formulated at the level of signed support masses. To interpret those trajectories as genuine probabilistic model trajectories, the probability layer must first satisfy more than the lightweight normalization contract built into the original `Model` structure. This section records the stronger layer and the verified lift from rational weight vectors to complete 4-PEL models.

9.1 Finite probability integrity

The compatibility predicate `FiniteProbabilityIntegrity` strengthens one local measure by requiring duplicate-free support, nonnegative event masses, extensionality with respect to event membership, monotonicity, disjoint finite additivity, empty-event mass zero, and total support mass one. The wrapper `StrongProbabilityModel` equips an ordinary 4-PEL model with this stronger certification.

The stronger contract is deliberately additive rather than a replacement of the legacy model interface. Earlier finite witnesses therefore remain available, while later geometric results can state explicitly when genuine finite-probability behavior is required.

Proposition 9.1 (Weight-generated finite probability)**.** *Let R be a duplicate-free finite support and let $q : R \to \mathbb{Q}$ be nonnegative with total mass one. Define the mass of a finite event S by summing the weights of the points of R that belong to S. Then the resulting local measure satisfies `FiniteProbabilityIntegrity`.* Status: Proved; trust inventory in Appendix A.

The implementation iterates over the fixed support rather than over the event list. Consequently event order and duplicate presentation do not alter the generated mass.

9.2 Convexity of the finite probability simplex

For two valid weight distributions q_0, q_1 on the same support and rational $t \in [0, 1]$, define

$$q_t(x) = (1 - t)q_0(x) + tq_1(x).$$

Lean verifies that q_t is again a valid finite weight distribution. Hence the full rational line segment between two valid endpoints remains in the finite probability simplex.

More strongly, every fixed event mass is affine in the same parameter:

$$\mu_t(S) = (1 - t)\mu_0(S) + t\mu_1(S).$$

Theorem 9.2 (Affine event mass on the probability simplex)**.** *For every fixed event S, the weight-generated mass along the convex distribution path satisfies*

$$\mu_{q_t}(S) = (1 - t)\mu_{q_0}(S) + t\mu_{q_1}(S).$$

STATUS: PROVED; TRUST INVENTORY IN APPENDIX A.

This identity is the formal bridge between the probability simplex and the affine scalar paths used in the threshold-crossing theorems.

9.3 A model-valued rational path

The probability path can be lifted to the complete model structure. Fix a world list W, an accessibility field R, an atomic valuation v, and Lockean thresholds c. Let q_0 and q_1 provide valid local weight distributions at every agent/world pair. For every rational $t \in [0, 1]$, define

$$M_t = (W, R, \mu_t, v, c),$$

where each local measure is generated from the interpolated weights q_t.

Theorem 9.3 (Convex strong model path)**.** *Every rational $t \in [0, 1]$ determines a `StrongProbabilityModel` M_t. Along the entire path,*

$$W_t = W, \qquad R_t = R, \qquad v_t = v, \qquad c_t = c,$$

and every fixed local event has affine probability mass. STATUS: PROVED; TRUST INVENTORY IN APPENDIX A.

Thus the geometric probability path is realized by complete probabilistically certified 4-PEL models, rather than merely by pairs of numbers in a support plane.

Exact scope of the realization. Two distinctions remain essential. First, this theorem concerns weight-generated strong endpoint models on a common semantic skeleton. It does not retroactively prove that every legacy witness in the repository satisfies the stronger probability contract. Second, a convex model path is not the same thing as an update-generated path: the theorem does not claim that every intermediate M_t arises by conditionalizing the source model on some evidence event.

There is also a formula-level boundary. The affine theorem applies directly to a *fixed event S*. For an arbitrary modal formula containing probabilistic belief, the set of worlds that positively or negatively support the formula can itself vary with t. In that case the event whose mass is measured is no longer fixed, so affine event mass alone does not imply affine `modalPositiveBeliefMass` or `modalNegativeBeliefMass`.

The next formal step is therefore narrower and sharper: identify a path-invariant formula fragment, beginning with atomic propositions and probability-free formulas, and lift the verified threshold-crossing and intermediate-phase geometry to those genuine model paths.

10 Structural Interpretation: From Reachability to Phase Kinematics

STATUS: STRUCTURAL INTERPRETATION.

The formal results support a layered interpretation of four-valued epistemic change. At the support-pair level, a belief formula is represented by two rational masses; the reliability refinement in Section 13 retains further information. Thresholding projects this point into one of four regions. The resulting FDE value forgets the precise probabilities and remembers only which side of the two Lockean walls each coordinate occupies.

This can be written as a coarse-graining map

$$q_c : \mathbb{Q}^2 \longrightarrow \{\mathbf{T}, \mathbf{F}, \mathbf{B}, \mathbf{N}\},$$

$$q_c(p^+, p^-) = ([p^+ \geq c], [p^- \geq c]).$$

The two threshold lines partition the support plane into four cells. The Boolean threshold square is the adjacency graph of these cells. In this sense the four-valued state space is not an arbitrary collection of labels. It is the quotient of a signed evidence space by two threshold predicates.

10.1 Global freedom, local constraint

Two verified results together give the central dynamic contrast:

$$\text{every ordered categorical transition is realizable,}$$

while

$$\text{every categorical change is mediated by a threshold-side change.}$$

Global reachability is therefore maximal, but local motion is not structureless. A transition leaves a trace in the signed support coordinates. Zero-wall movement is robust, one-wall movement crosses one categorical boundary, and two-wall movement crosses both.

The affine results sharpen this further. A diagonal transition does not merely differ in both bits. Along the chosen support interpolation it contains two unique wall events. Those events may coincide, or they may be temporally ordered. If they are ordered, a forced adjacent phase lies between them. The endpoint displacement therefore hides a short event history.

It is useful to call this structure *epistemic phase kinematics*. The term is not an additional theorem. It names the combination of four verified layers:

$$\text{cell membership,} \quad \text{wall count,} \quad \text{crossing time,} \quad \text{intermediate phase sequence.}$$

10.2 The exceptional role of simultaneity

For a diagonal move, sequential crossings force an intermediate adjacent phase. Simultaneity removes the nonempty open interval between the two wall hits, although an isolated boundary value can still occur. Geometrically, simultaneity requires passage through (c, c). This is an alignment condition, not a verified measure-theoretic claim about generic trajectories.

This distinction may become important in later model-level dynamics. If admissible model paths can realize the affine support interpolation, sequential diagonal changes would carry genuine intermediate epistemic states, while simultaneous changes would behave as boundary-coincident transitions.

10.3 Connection to structural transport

A broader working hypothesis in the 4-PEL project treats epistemic paradoxes as failures of structural transport. The present dynamic results fit that program in a precise but limited way. Thresholding is a projection from rich signed probabilistic information to a coarse four-valued state. Endpoint states alone do not preserve crossing order. Recovering the order requires returning to the richer support trajectory.

The paper does not claim that all epistemic paradoxes reduce to threshold crossing. Rather, this case study supplies one clean instance of a recurring pattern:

$$\text{rich structure} \longrightarrow \text{coarse projection} \longrightarrow \text{apparent categorical jump},$$

where restoring the discarded structure reveals the mechanism hidden by the projection.

The formal contribution of the present paper ends before this philosophical generalization. A general structural-transport theorem connecting threshold dynamics to the project's other paradox families remains open.

11 Lean Mechanization and Verification Status

The artifact uses Lean 4.31.0 without Mathlib. The current branch is `research/modal-probabilistic-fine-graining`; the root module `PEL4.lean` imports the relevant development. Appendix A provides the claim-to-declaration map, including the reliability refinement. The explicit audit files are the machine-readable list of selected claims; this selection is not a certification of every declaration in the repository.

11.1 Verification contract

A successful build establishes that declarations typecheck, not that all their assumptions are proved. Version 0.4 therefore audits 52 manuscript declarations in `PEL4/PaperAxiomAudit.lean` and 24 MPFG declarations in its separate audit. The full build and both strict audits passed for source revision `07fbc5193e0f7e43de565158c0eaf0cbb9c08dfb`; the review report records the CI run and exact dependencies. Only `propext`, `Classical.choice`, and `Quot.sound`, or subsets thereof, occur in these selected chains. `sorryAx`, project-specific axioms, and native-decision axioms are rejected.

The review found that some general headlines inherited native-evaluation assumptions through elementary helpers. In eight source modules, the relevant finite computations were changed from `native_decide` to `decide+kernel`, without changing statements or semantic definitions. Legacy declarations outside the audited chains may still use native evaluation or project axioms; no repository-wide axiom-freedom claim is made.

11.2 Status discipline

The manuscript follows the repository verification policy:

(1) **PROVED**: the stated theorem typechecks and its audited dependency chain contains no project-specific unproved assumption. Explicit hypotheses and model proof fields remain assumptions of the theorem's quantified statement.

(2) **FINITE-WITNESS**: a finite construction or counterexample is verified. Finite domain coverage is not a theorem about arbitrary model sizes.

(3) **EXPERIMENTAL / INTERPRETIVE**: terminology, philosophical readings, and prospective extensions without a corresponding established theorem.

(4) **AXIOMATIC-PROTOTYPE**: a substantive property is postulated. No manuscript headline is promoted from this category merely because it compiles.

The proof sketches explain the main arguments for readers; Lean verifies their formal counterparts. The geometry uses rational algebra, not topological continuity. Numerical figure coordinates illustrate those statements and are not additional model-realization certificates.

Reproducibility. From a clean clone, or after preserving local work:

```
git fetch origin
git switch research/modal-probabilistic-fine-graining
git pull --ff-only
lake build
python3 scripts/check_mpfg_axioms.py
python3 scripts/check_paper_axioms.py --report paper-axioms.json
python3 -m unittest discover -s scripts -p 'test_*.py'
cd paper
latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex
```

The mutable branch is for ongoing development; use a recorded commit for an exact reproduction. Lean checking and manuscript rendering remain separate checks. The committed PDF is the rendered working artifact, not evidence of formal correctness by itself.

12 Related Work, Limits, and Open Problems

The semantic kernel belongs to the Belnap–Dunn/FDE tradition of tracking support for truth and falsity independently [1, 5]. Four-valued modal logics with bilateral relational semantics have been studied in several forms [8, 10]. The present project adds a finite probabilistic threshold layer and a primitive evidence-stable knowledge operator whose behavior is analyzed mechanically rather than assumed to coincide with a standard modal box.

The threshold connection between all-or-nothing belief and degrees of belief belongs to the broader Lockean tradition. Leitgeb's stability theory is a particularly important point of comparison because it places stability conditions at the interface between probability and categorical belief [7]. The 4-PEL notion of stability is different in type: it concerns invariance of the complete FDE value across an accessible profile and acts as a filter from threshold belief to knowledge. A careful conceptual comparison is therefore needed before stronger claims about similarity or difference are made.

The update perspective is adjacent to dynamic epistemic logic, where informational actions transform epistemic models and knowledge states [11]. The conditionalization operator studied here is narrower: it preserves worlds, accessibility, valuation, and threshold and changes only local probabilities. Conversely, the four-valued signed-support geometry developed here is not part of the standard DEL vocabulary. The manuscript reports this construction without claiming that it is novel relative to all probabilistic or nonclassical dynamic epistemic systems.

Direct probability predecessors. Klein, Majer, and Rafiee Rad [6] already develop probabilities over Belnap–Dunn logic, including axiomatization and Bayesian and Jeffrey-style updating. Thus neither probabilistic gaps/gluts nor their updating is introduced by the present paper. Bílková et al. [2] explicitly compare signed probabilities with the four-region account and give two-layered logical systems. Their languages and representation results are important comparators, not theorems inherited by our recursively nested threshold language. No translation from their full logics to `ModalFormula` is proved here.

Reliability and status modalities. Coniglio and Rodrigues [4] provide six-valued logics with classicality propagation; their six-scenario motivation predates the 2026 probability preprint of Borja Macias, Coniglio, and Hernández-Tello [3]. The latter's abstract announces probability and Jeffrey-update results over that logic. MPFG does not reproduce these results: its tags are given data, it has no classicality connective or truth tables, and it has not yet defined reliability-sensitive updates. Likewise, Petrukhin [9] gives four-valued essence/accident modalities and S5-based proof systems. Our ordinary meta-level status predicates are not those object-language operators or their completeness theorems.

This is a targeted positioning check, not an exhaustive novelty search. The comparisons use the published descriptions and, for the two-layered and six-valued works, the accessible preprint text. The 2026 probability work is used at abstract level only; no detailed equivalence or priority judgment depends on an unavailable full-text comparison.

Novelty boundary. No strong novelty claim is made in this version. The individual mathematical ingredients are elementary: Boolean pairs, threshold comparisons, affine interpolation, finite convex probability, linear order, and midpoint arguments. The research contribution currently established internally to the project is the machine-checked composition of these ingredients into one dependency chain connecting admissible probabilistic update to four-valued categorical reachability, robustness, crossing order, intermediate phase structure, and a model-valued rational probability path. A systematic literature audit is publication-critical.

12.1 Verified model-path layer and remaining realization questions

The earlier support-to-model existence problem now has a positive answer for a precise class. Given two weight-generated strong endpoint models on the same semantic skeleton, rational convex interpolation of their local weights produces a `StrongProbabilityModel` for every $t \in [0,1] \cap \mathbb{Q}$. Worlds, accessibility, valuation, and threshold remain fixed, and the probability of every fixed event varies affinely.

Two stronger realization questions remain open.

Research Question 12.1 (Formula-level support realization)**.** For which modal formulas φ are the positive and negative support events invariant along a convex strong model path, so that the verified affine fixed-event theorem applies directly to $P_t^+(\varphi)$ and $P_t^-(\varphi)$?

Atomic propositions are the natural first case because their valuation is fixed by construction. A broader probability-free fragment is plausible but requires an explicit theorem connecting path-invariance of `evalModal` to fixed support-event membership.

Research Question 12.2 (Update-generated path)**.** When can the intermediate models of a convex strong model path themselves be obtained by admissible conditionalization of the source model, possibly using a parameterized family of evidence events?

A model-valued path and an update-generated path are different notions. The convex-model-path module verifies the former and deliberately does not infer the latter. On a fixed finite universe, event-only conditioning of one fixed prior yields only finitely many posteriors (one per positive-mass event). A nonconstant affine weight segment, in contrast, has infinitely many rational points. Hence a general identification cannot hold without changing the update mechanism. This elementary obstruction is a mathematical observation here, not an additional Lean theorem. Soft evidence or likelihood updates would require a separately specified semantics.

12.2 Open problem 2: arbitrary continuous paths

The affine crossing theorem is constructive and sufficient for the present phase result, but it is not a general topological theorem. A stronger layer would define a continuous support path

$$\gamma : [0,1] \to \mathbb{R}^2$$

and prove that endpoint straddling forces intersection with the appropriate threshold wall. Standard intermediate-value reasoning would then recover a crossing theorem independent of linear interpolation.

For two-wall transitions, arbitrary paths introduce genuinely new geometry. A path may meet the two walls multiple times, touch a wall without changing side, or revisit a categorical cell. The current uniqueness and three-case temporal order are specific to nonconstant affine coordinates. A general theory would therefore need a richer notion of crossing event than the present unique hit parameter.

12.3 Open problem 3: algebraic stability

The knowledge filter is presently defined semantically by full-value homogeneity over an accessible range. An open problem is to characterize this stability intrinsically in bilattice or algebraic terms and to determine which logical operators preserve, reflect, or destroy it. Such a characterization could connect the dynamic phase table to existing algebraic semantics for four-valued modal systems.

12.4 Open problem 4: higher-dimensional phase complexes

A single belief formula contributes two threshold bits. For n tracked formulas, the naive categorical state space has $2n$ threshold coordinates and therefore resembles a Boolean hypercube. This suggests a higher-dimensional generalization of wall count, crossing order, and phase sequence. Dependencies between formulas, however, can identify or forbid regions of that cube. Determining the realizable cell complex is a natural extension of the present two-dimensional result.

12.5 Open problem 5: structural transport and paradox families

The project contains independent formal diagnoses of Lottery, Preface, Sorites, Knower, Surprise Examination, Moore/Fitch, and dynamic threshold phenomena. The working phrase "paradoxes as failures of structural transport" is currently interpretive. A genuine general theorem would require an abstract account of transformations, observations, preserved predicates, and counterexamples that subsumes several of these families without erasing their differences.

13 Modal-probabilistic refinement and information loss

This extension is a semantic refinement of evidence, not a replacement of the four-valued algebra. Its formal declarations are in `PEL4/ModalProbability/Refinement.lean` and `PEL4/ModalProbability/StatusTransport.lean`. Here "conservative" means preservation of the specified observations under projection, not proof-theoretic conservativity of a new calculus. The gate's validation status is recorded in `docs/MODAL_PROBABILISTIC_FINE_GRAINING.md`.

Definition 13.1 (Reliability-tagged evidence profile)**.** A fine profile is a nonnegative rational vector $q = (t, b, n, f, t_r, f_r)$ of total mass one. Its coarse projection is

$$\pi(q) = (t + t_r, b, n, f + f_r).$$

The reliability tags are supplied semantic data, not proofs of empirical truth. Untagged does not mean unreliable. The six cells select a subset of the earlier unrestricted `EvidenceStatus`, which also permits other tag combinations; they do not classify that whole type. The untagged lift of (t, b, n, f) is $(t, b, n, f, 0, 0)$.

Proposition 13.2 (Affine projection with a section)**.** *The untagged lift is a right inverse of* π*. Signed support is preserved:* $P^+(q) = t + t_r + b$ *and* $P^-(q) = f + f_r + b$*. For* $0 \le a \le 1$*, both* *normalized rational simplexes are closed under mixing, and the formal projection identity is*

$$\pi((1-a)q + ar) = (1-a)\pi(q) + a\pi(r).$$

Thus raw threshold belief commutes with forgetting reliability. This does not assert that thresholding is affine; the raw belief compatibility follows from the chosen definitions. Moreover, no function of the coarse profile recovers reliable mass $t_r + f_r$ in general: pure untagged truth and pure reliable truth have the same projection, but reliable masses zero and one. Status: Proved; trust inventory in Appendix A.

Proof sketch. Summing the two truth-tag coordinates and the two falsity-tag coordinates commutes with coordinatewise convex mixing by distributivity. The untagged lift sets both tag masses to zero. If reliability were a function of the coarse profile, the two pure-truth profiles would force its value to be both zero and one. □

Definition 13.3 (Status stability and local fibre rigidity)**.** For any relation R, observation v, and projection π, put

$$\begin{aligned} \mathrm{Stable}_{\mathrm{cur}}(v, w) &\iff \forall u\,(Rwu \Rightarrow v(u) = v(w)), \\ \mathrm{Rigid}(v, \pi, w) &\iff \forall u\,(Rwu \Rightarrow (\pi(v(u)) = \pi(v(w)) \Rightarrow v(u) = v(w))). \end{aligned}$$

The subscript distinguishes current-world stability from the accessible-range homogeneity H used by primitive K. On reflexive frames they agree for the same observation, but no such identification is used on arbitrary frames.

Theorem 13.4 (Local stability decomposition)**.** *For any observation and projection, with no frame assumptions,*

$$\mathrm{Stable}_{\mathrm{cur}}(v, w) \iff \mathrm{Stable}_{\mathrm{cur}}(\pi \circ v, w) \wedge \mathrm{Rigid}(v, \pi, w).$$

Status: Proved; trust inventory in Appendix A.

Proof. Fine stability implies both coarse stability (apply π) and rigidity. Conversely, coarse stability puts every accessible observation in the current coarse fibre, and rigidity then identifies its fine value with the current one. □

This is an elementary local decomposition, not a test that guarantees rigidity independently. It does not assert that π is globally injective. Two universally accessible worlds carrying untagged and reliable truth witness coarse stability with fine instability, even on a reflexive, transitive, Euclidean frame. A second witness distinguishes mass stability from categorical stability: $(0, 1, 0, 0)$ and $(0, 3/4, 1/4, 0)$ both yield $\mathbf{B}$ at threshold $3/5$, although their glut masses differ.

For a status s, the auxiliary predicates are

$$\mathrm{Ess}_s(w) \iff (v(w) = s \Rightarrow \forall u\,(Rwu \Rightarrow v(u) = s)),$$

$$\mathrm{Acc}_s(w) \iff v(w) = s \wedge \exists u\,(Rwu \wedge v(u) \neq s).$$

Thus Ess_s may hold vacuously when the current value is not s. Its accidental counterpart requires an actual different accessible status. These are classical meta-level predicates, not additional FDE connectives. Their relationship to the existing executable stability test and K is stated separately: applying the unchanged knowledge interpretation to raw fine belief or to its coarse projection yields the same value. No identification of K with ordinary S5 necessity is made.

Finally, the existing finite-step-chain theorem transports threshold status when each edge preserves it. Cell refinement is distinct from population Finite Fine-Grainedness, and connectivity alone does not imply status preservation. Normalized reliability-sensitive updates, probability-one versus necessity under full-support assumptions, and any representation of proof-theoretic fractional values remain subsequent research targets.

14 Conclusion

The case study establishes three bounded results. First, the stipulated evidence-stable knowledge operator factors through accessible-profile homogeneity; probability-only conditionalization realizes all sixteen $\mathsf{K}(\mathsf{B}p)$ endpoint values across a six-world witness family. This is not a claim that one fixed model realizes every update, nor an empirical theory of knowledge.

Second, signed thresholding reduces quantitative support to two Boolean decisions. On a chosen rational affine segment, straddling gives unique wall-hit parameters and their order determines a sequential midpoint phase. An actual discrete update does not supply this interpolated history. At simultaneous hits the inclusive threshold yields **B**, which may occur only at the boundary parameter.

Separately, normalized rational world weights construct probability-integrity-certified model paths with affine fixed-event masses. Connecting these paths to the supports of arbitrary probability-sensitive formulas, or generating them by admissible updates, remains unproved. The weakest model interface is not silently promoted to a finite probability space.

Third, six-cell reliability tagging preserves coarse support and convex mixing but loses recoverable reliability under projection. Fine current-world stability holds exactly when coarse stability and local fibre rigidity both hold. The S5-frame witnesses show that frame geometry alone cannot restore discarded information. The tags are supplied semantic data, not a derivation of reliability or a six-valued logical calculus.

The contribution is a reproducible mechanized analysis of preservation and information loss. The declaration-level audits, explicit finite witnesses, and proof sketches identify what is established and under which contract. Before journal submission, the most valuable next steps are a strong certificate for the reachability family, a formula-level model-path bridge, and a theorem-by-theorem comparison with the probability and evidence-logics literature. No claim of completeness, topological generality, or priority is inferred from a successful build.

A Formal Correspondence

This appendix maps the principal mathematical claims in the manuscript to the Lean modules and theorem names that support them. The list is selective rather than exhaustive. All names below are in namespace `PEL4`; the audit files enumerate exact fully qualified targets. A theorem's hypotheses remain part of its claim.

Paper claim	Lean theorem / definition	Module
Four FDE values as bilateral Boolean pairs	`FDEValue`, `FDEValue.T/F/B/N`	`PEL4/FDE.lean`
Signed threshold belief	`belief`	`PEL4/Belief.lean`
Primitive modal evaluator with $\mathsf{B}, \mathsf{K}, \Diamond$	`ModalFormula`, `evalModal`	`PEL4/ModalLanguage.lean`
Knowledge equals stability-filtered belief	`modal_knowledge_as_stability_filtered_belief`	`PEL4/ModalKnowledgeBeliefFactorization.lean`
Safe probability-only update	`ConditionalizationAdmissible`, `conditionalize`	`PEL4/Dynamics.lean`
Probability-free invariance	`evalModal_conditionalize_of_probabilityFree`	`PEL4/ModalDynamicsStability.lean`

Four dynamic stability phases	`modal_knowledge_phase_stable_stable`, `modal_knowledge_phase_stable_unstable`, `modal_knowledge_phase_unstable_stable`, `modal_knowledge_phase_unstable_unstable`	`PEL4/ModalDynamicsPhaseClassification.lean`
Both stability-change directions are realizable	`conditionalization_realizes_both_stability_phase_directions`	`PEL4/ModalDynamicsPhaseClassification.lean`
All sixteen $\mathsf{K}(\mathsf{B}p)$ endpoint transitions	`conditionalization_realizes_every_knowledge_value_transition`	`PEL4/ModalDynamicsReachability.lean`
Creation of a knowledge glut	`conditionalization_can_create_knowledge_glut`	`PEL4/ModalDynamicsReachability.lean`
Glut-to-gap and gap-to-truth transitions	`conditionalization_can_turn_knowledge_glut_into_gap`, `conditionalization_can_turn_knowledge_gap_into_truth`	`PEL4/ModalDynamicsReachability.lean`
Exact belief robustness boundary	`modal_belief_value_eq_iff_threshold_bits_eq`	`PEL4/ModalDynamicsRobustness.lean`
Compositional robust formulas are invariant	`evalModal_conditionalize_of_robust`	`PEL4/ModalDynamicsRobustness.lean`
Robustness strictly extends the probability-free fragment	`diagonal_reachability_belief_is_robust`, `diagonal_reachability_belief_not_probabilityFree`	`PEL4/ModalDynamicsRobustness.lean`
Threshold wall count	`thresholdWallCount`	`PEL4/ModalDynamicsGeometry.lean`
Zero wall iff equal FDE state	`thresholdWallCount_eq_zero_iff`	`PEL4/ModalDynamicsGeometry.lean`
Two walls iff both coordinates flip	`thresholdWallCount_eq_two_iff`	`PEL4/ModalDynamicsGeometry.lean`
Wall flip iff endpoint straddling	`threshold_decision_ne_iff_straddles`	`PEL4/ModalDynamicsThresholdCrossing.lean`
Two walls iff both supports straddle	`conditionalization_belief_two_walls_iff_both_supports_straddle`	`PEL4/ModalDynamicsThresholdCrossing.lean`
Affine path	`affineRatPath`	`PEL4/ModalDynamicsAffineCrossing.lean`
Straddling gives unit-interval threshold hit	`thresholdStraddles_has_affine_unit_crossing`	`PEL4/ModalDynamicsAffineCrossing.lean`
Every belief change has an affine wall hit	`conditionalization_belief_change_has_affine_threshold_crossing`	`PEL4/ModalDynamicsAffineCrossing.lean`
Two-wall transition has two affine hits	`conditionalization_belief_two_walls_has_two_affine_crossings`	`PEL4/ModalDynamicsAffineCrossing.lean`
Unique affine threshold time	`affineRatPath_threshold_hit_unique`	`PEL4/ModalDynamicsCrossingOrder.lean`

Crossing-order trichotomy	`affineCrossingPair_order_trichotomy`	`PEL4/ModalDynamicsCrossingOrder.lean`
Simultaneous iff common (c, c) hit	`affineCrossingPair_simultaneous_iff_common_hit`	`PEL4/ModalDynamicsCrossingOrder.lean`
Conditionalized two-wall order classification	`conditionalization_belief_two_walls_crossing_order`	`PEL4/ModalDynamicsCrossingOrder.lean`
Midpoint lies between sequential crossings	`affineCrossingMidpoint_between_positive_first`, `affineCrossingMidpoint_between_negative_first`	`PEL4/ModalDynamicsIntermediatePhase.lean`
Source side before and target side after crossing	`affine_threshold_side_before_crossing_eq_source`, `affine_threshold_side_after_crossing_eq_target`	`PEL4/ModalDynamicsIntermediatePhase.lean`
Positive-first midpoint state	`affineCrossingPair_positive_first_midpoint_state`	`PEL4/ModalDynamicsIntermediatePhase.lean`
Negative-first midpoint state	`affineCrossingPair_negative_first_midpoint_state`	`PEL4/ModalDynamicsIntermediatePhase.lean`
Intermediate is adjacent to both diagonal endpoints	`positiveFirstIntermediate_adjacent_to_diagonal_endpoints`, `negativeFirstIntermediate_adjacent_to_diagonal_endpoints`	`PEL4/ModalDynamicsIntermediatePhase.lean`
Concrete eight-case diagonal table	`affine_diagonal_intermediate_vertex_table`	`PEL4/ModalDynamicsIntermediatePhase.lean`
Full simultaneous/sequential intermediate-phase theorem	`conditionalization_belief_two_walls_intermediate_phase`	`PEL4/ModalDynamicsIntermediatePhase.lean`
Strong finite-probability contract	`FiniteProbabilityIntegrity`, `StrongProbabilityModel`	`PEL4/FiniteProbabilityIntegrity.lean`
Weight-generated measure satisfies integrity	`weightGeneratedMeasure_integrity`	`PEL4/WeightGeneratedProbability.lean`
Convex interpolation preserves distributions	`convexWeightDistribution`	`PEL4/ConvexProbabilitySimplex.lean`
Every fixed event mass is affine	`weightedEventMass_convex`	`PEL4/ConvexProbabilitySimplex.lean`
Complete strong rational model path	`convexStrongModelAt`	`PEL4/ConvexModelPath.lean`
Worlds, accessibility, valuation, and threshold fixed	`convexStrongModelAt_worlds`, `convexStrongModelAt_accessibility`, `convexStrongModelAt_valuation`, `convexStrongModelAt_threshold`	`PEL4/ConvexModelPath.lean`
Fixed-event mass affine along full model path	`convexStrongModelAt_eventMass`	`PEL4/ConvexModelPath.lean`
Nonempty accessibility and fixed-input K invariance	`PaperReview.accessibility_nonempty`, `PaperReview.knowledge_ignores_probability_for_fixed_values`	`PEL4/PaperReviewBoundaries.lean`

Inclusive common hit is B; simultaneous T/F witness	`PaperReview.threshold_intersection_is_B`, `PaperReview.simultaneous_T_F_has_B_hit`	`PEL4/PaperReviewBoundaries.lean`
Six-cell section and nonrecoverable reliability	`ModalProbability.SixCellProbability.coarse_untagged`, `ModalProbability.SixCellProbability.no_reliability_reconstruction`	`PEL4/ModalProbability/Refinement.lean`
Affine coarse projection	`ModalProbability.coarse_mix`	`PEL4/ModalProbability/Refinement.lean`
Local fine/coarse stability boundary	`ModalProbability.stable_iff_coarse_and_fibre_rigid`	`PEL4/ModalProbability/StatusTransport.lean`
S5-frame reliability and changing glut-mass witnesses	`ModalProbability.probability_profile_projection_hides_instability`, `ModalProbability.stable_B_does_not_mean_constant_mass`	`PEL4/ModalProbability/StatusTransport.lean`

Verification note. The full Lean 4.31 CI build and strict audits described in Section 11 passed. The audited manuscript and MPFG declarations have only standard Lean axiom dependencies. Their statements, not informal paraphrases, determine the precise scope of verification. The formula-level lift from fixed events to arbitrary probability-sensitive support events remains separate and is not inferred from the model-path theorem.

References

[1] Nuel D. Belnap. A useful four-valued logic. In J. Michael Dunn and George Epstein, editors, *Modern Uses of Multiple-Valued Logic*, pages 5–37. D. Reidel, 1977. `https://doi.org/10.1007/978-94-010-1161-7_2`.

[2] Marta Bílková, Sabine Frittella, Daniil Kozhemiachenko, and Ondrej Majer. Two-layered logics for probabilities and belief functions over Belnap–Dunn logic. *Mathematical Structures in Computer Science*, 35:1–35, 2025. Article e1. `https://doi.org/10.1017/S0960129525000064`.

[3] Verónica Borja Macias, Marcelo E. Coniglio, and Alejandro Hernández-Tello. Probabilities beyond Belnap-Dunn logic: dealing with gaps, gluts and reliability, 2026. Preprint, arXiv:2608.20228v1, 20 August 2026. `https://arxiv.org/abs/2608.20228`.

[4] Marcelo E. Coniglio and Abilio Rodrigues. From Belnap-Dunn four-valued logic to six-valued logics of evidence and truth. *Studia Logica*, 112:561–606, 2024. Published online 2023; preprint arXiv:2209.12337 (2022). `https://doi.org/10.1007/s11225-023-10062-5`.

[5] J. Michael Dunn. Intuitive semantics for first-degree entailments and 'coupled trees'. *Philosophical Studies*, 29:149–168, 1976. `https://doi.org/10.1007/BF00373152`.

[6] Dominik Klein, Ondrej Majer, and Soroush Rafiee Rad. Probabilities with gaps and gluts. *Journal of Philosophical Logic*, 50(5):1107–1141, 2021. `https://doi.org/10.1007/s10992-021-09592-x`.

[7] Hannes Leitgeb. The stability theory of belief. *The Philosophical Review*, 123(2):131–171, 2014.

[8] Sergei P. Odintsov and Heinrich Wansing. Modal logics with belnapian truth values. *Journal of Applied Non-Classical Logics*, 20(3):279–301, 2010.

[9] Yaroslav Petrukhin. Essence and accident modalities meet belnapian truth values. *Studia Logica*, 2026. Published online 15 July 2026. `https://doi.org/10.1007/s11225-026-10250-z`.

[10] Umberto Rivieccio, Achim Jung, and Ramon Jansana. Four-valued modal logic: Kripke semantics and duality. *Journal of Logic and Computation*, 27:155–199, 2017. Published online 2015. `https://doi.org/10.1093/logcom/exv038`.

[11] Hans van Ditmarsch, Wiebe van der Hoek, and Barteld Kooi. *Dynamic Epistemic Logic*, volume 337 of *Synthese Library*. Springer, Dordrecht, 2007.